IPv4 Subnetting Made Simple

A beginner-friendly lesson with worked examples, formulas, and practice questions

Lesson goal
Learn how CIDR prefixes, subnet masks, block sizes, network addresses, and broadcast addresses work together.

What you will learn

- What subnetting is and why networks are divided into subnets
- How to interpret IPv4 addresses and CIDR notation
- How to convert a CIDR prefix into a subnet mask
- How to calculate total addresses, usable hosts, and block size
- How to find network and broadcast addresses
- How to split a /27 network into four /29 subnets

Keep this rule in mind: the network address is the first address in a subnet; the broadcast address is the last address in a traditional IPv4 subnet.

1. What is subnetting?

Subnetting divides one IP network into smaller logical networks. Think of a large apartment building divided into sections. In a company, different subnets might be used for employee computers, servers, guest Wi-Fi, or security cameras.

Original network **10.200.20.0/27** **32 total addresses**	
↓ Split into four equal subnets	
Subnet 1 10.200.20.0/29	**Subnet 2** 10.200.20.8/29
Subnet 3 10.200.20.16/29	**Subnet 4** 10.200.20.24/29

2. Understand IPv4 and CIDR

An IPv4 address has four octets separated by periods, such as 10.200.20.0. Each octet ranges from 0 to 255, and the full address contains 32 bits. A bit is a 0 or 1.

CIDR (Classless Inter-Domain Routing) uses a slash number to show how many bits belong to the network portion. In 10.200.20.0/27, 27 bits are network bits and 32 – 27 = 5 bits remain for hosts.

3. CIDR and subnet mask quick reference

/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

*For ordinary IPv4 subnets, usable hosts are usually total addresses minus two: one network address and one broadcast address.

4. How to convert /27 to a subnet mask

A /27 has 27 network bits: 24 fill the first three octets, and 3 more fill the last octet. In binary, the mask is 11111111.11111111.11111111.11100000.

Binary octet	Decimal
11111111	255
11111111	255

11111111	255
11100000	224

Result: 255.255.255.224

5. Network, host, and broadcast addresses

For a traditional IPv4 subnet, remember these three concepts:

Network address	First address; identifies the subnet.	10.200.20.0
Usable host range	Addresses normally assigned to devices.	10.200.20.1 – 10.200.20.30
Broadcast address	Last address; reaches all hosts on the subnet.	10.200.20.31

6. The block-size shortcut

Block size = 256 – the subnet-mask value in the changing octet. For a /27, the mask is 255.255.255.224, so the last octet is the changing octet.

256 – 224 = 32

The network addresses increase by 32: 10.200.20.0, then 10.200.20.32, then 10.200.20.64, and so on. The broadcast address is one less than the next network address. For the first block, it is 10.200.20.31.

7. Split 10.200.20.0/27 into four subnets

Step 1: To create 4 subnets, borrow 2 bits because $2^2 = 4$. Step 2: Add those bits to the prefix: /27 + 2 = /29. Step 3: A /29 has 3 host bits, so $2^3 = 8$ total addresses per subnet. Step 4: Count by 8.

1st	10.200.20.0/29	.1 – .6	10.200.20.7
2nd	10.200.20.8/29	.9 – .14	10.200.20.15
3rd	10.200.20.16/29	.17 – .22	10.200.20.23
4th	10.200.20.24/29	.25 – .30	10.200.20.31

Exam reminder: the original /27 broadcast is 10.200.20.31. The second new /29 subnet broadcast is 10.200.20.15. Always check whether the question refers to the original network or a subnet created from it.

8. Formula cheat sheet

What you need	Formula / rule
Total addresses	2^(32 – CIDR prefix)
Usable hosts (typical IPv4 subnet)	2^(32 – prefix) – 2
Number of subnets created	2^(bits borrowed)
New CIDR prefix	Original prefix + borrowed bits
Block size	256 – mask value in the changing octet

9. Practice exam

1. What is the decimal subnet mask for /27?

A. 255.255.255.0 B. 255.255.255.192 C. 255.255.255.224 D. 255.255.255.240

2. How many total addresses are in a /29 subnet?

A. 4 B. 8 C. 16 D. 32

3. What is the broadcast address of 10.200.20.0/27?

A. 10.200.20.30 B. 10.200.20.31 C. 10.200.20.32 D. 10.200.20.255

4. Split 10.200.20.0/27 into 4 equal subnets. What is subnet 3's network address?

A. 10.200.20.8 B. 10.200.20.12 C. 10.200.20.16 D. 10.200.20.24

5. What is the broadcast address of the second /29 subnet?

A. 10.200.20.8 B. 10.200.20.14 C. 10.200.20.15 D. 10.200.20.16

Try answering all five before checking the key on the next page.

10. Answer key and explanations

1. C — 255.255.255.224

A /27 has 27 network bits. The last octet is 11100000, which equals 224.

2. B — 8 addresses

32 – 29 = 3 host bits; $2^3 = 8$ total addresses.

3. B — 10.200.20.31

A /27 block contains .0 through .31, so .31 is the broadcast.

4. C — 10.200.20.16

The new prefix is /29 and the block size is 8. Network addresses are .0, .8, .16, and .24.

5. C — 10.200.20.15

The second /29 subnet spans .8 through .15, so .15 is the broadcast.

What to memorize first

1. /27 = 255.255.255.224
2. Block size = 256 – mask value
3. Network address = first address
4. Broadcast address = last address
5. Borrow bits to create more subnets